Periodic viscous Burgers as a positive projective heat flow

Global phase portrait, first-mode asymptotics, and a Route-A boundary

HCS-C195 research artifact

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Abstract

For every viscosity $\nu > 0$, circumference $L > 0$, mean $m \in \mathbb{R}$, and $s > 3/2$, we identify the periodic viscous-Burgers semiflow on its fixed-mean H^s leaf with the strictly positive projectivization of a drift–heat semigroup. This gives a global physical-time solution and instant smoothing. Every orbit converges to the unique constant equilibrium, so no nonconstant periodic or recurrent point exists. The first active Fourier mode of the positive lift gives an exact leading expansion, and the entire linearized spectrum is explicit.

1 Frozen phase leaf and the correct linear state

On $\mathbb{T}_L = \mathbb{R}/L\mathbb{Z}$, consider

$$u_t + uu_x = \nu u_{xx}, \qquad X_m^s = \{u \in H^s(\mathbb{T}_L;\mathbb{R}) : L^{-1}\int_{\mathbb{T}_L} u\,dx = m\}. \tag{1}$$

The clock is the source physical time. Define

$$\mathbb{P}_+^{s+1} = \{w \in H^{s+1}(\mathbb{T}_L;\mathbb{R}) : \min w > 0\}/\mathbb{R}_{>0}, \qquad \Phi_m([w]) = m - 2\nu\partial_x \log w. \tag{2}$$

The quotient is essential: positive scalar multiples give the same Burgers state. If $V_x = u - m$ and V is the unique zero-mean periodic primitive, then

$$\Phi_m^{-1}(u) = [e^{-V/(2\nu)}]. \tag{3}$$

Sobolev composition makes (2)–(3) smooth inverse maps for $s > 3/2$.

In fixed x, the autonomous linear semigroup is

$$K_t = e^{t(\nu\partial_x^2 - m\partial_x)}, \qquad K_t w(x) = \left(e^{\nu t\partial_x^2} w\right)(x - mt). \tag{4}$$

Thus pure heat appears after the Galilean coordinate $y = x - mt$; it is not mislabelled as the autonomous fixed-coordinate flow.

Theorem 1 (global projective Cole–Hopf conjugacy)**.** *For all parameters above,*

$$S_t\Phi_m([w]) = \Phi_m([K_t w]), \qquad t \geq 0. \tag{5}$$

The cone is invariant, every Burgers orbit is global, and $S_t u$ is smooth for every $t > 0$. Moreover $S_t u \to m$ in H^s. Hence $u \equiv m$ is the unique equilibrium, periodic point, and recurrent point on X_m^s, where recurrence means $S_{t_j} u \to u$ for some $t_j \to \infty$.

Write $\kappa_k = 2\pi k/L$ and $w_0 = \sum a_k e^{i\kappa_k x}$. If $u_0 \neq m$, let r be the first active absolute mode and r_2 the next one (or ∞). Then, with $\eta = \min(2\nu\kappa_r^2, \nu\kappa_{r_2}^2) > \nu\kappa_r^2$,

$$S_t u_0 - m = -\frac{2\nu}{a_0}\sum_{|k|=r} i\kappa_k a_k e^{-\nu\kappa_r^2 t} e^{i\kappa_k(x-mt)} + O_{H^s}(e^{-\eta t}). \tag{6}$$

Consequently the exact logarithmic decay exponent is $\nu\kappa_r^2$. The complexified linearization at m, with domain $H^{s+2} \subset H^s$, has complete spectrum

$$\lambda_k = -\nu\kappa_k^2 - im\kappa_k, \qquad k \in \mathbb{Z}, \tag{7}$$

with $k = 0$ removed on the fixed-mean tangent space.

2 Proof mechanism

If $w_t = \nu w_{xx} - m w_x$ and $u = m - 2\nu w_x/w$, direct differentiation gives $u_t + uu_x - \nu u_{xx} = 0$; conversely (3) recovers the same projective linear orbit. The periodic heat kernel preserves strict positivity, proving (5) and global smoothing.

Fourier diagonalization gives

$$K_t w_0 = a_0 + \sum_{k\neq 0} a_k e^{-\nu\kappa_k^2 t} e^{i\kappa_k(x-mt)}, \qquad a_0 > 0. \tag{8}$$

Hence $K_t w_0 \to a_0$ in H^{s+1}, so $S_t u_0 \to m$ in H^s. A returning subsequence must have both limits u and m, proving the recurrence claim. Separating the $|k| = r$ pair in (8), the next linear tail decays at $\nu\kappa_{r_2}^2$, while the first nonlinear term in $(a_0 + h)^{-1}$ decays at $2\nu\kappa_r^2$. This proves (6); its leading pair is nonzero and translation preserves its H^s norm. Finally $h_t = \nu h_{xx} - m h_x$ diagonalizes in Fourier modes. Compact resolvent gives the completeness asserted in (7).

References

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[2] J. D. Cole, *On a quasi-linear parabolic equation occurring in aerodynamics*, Quart. Appl. Math. 9 (1951), 225–236. JSTOR 43633894.